

Nuvoton MCU Evaluation Pre-Requirement

Nuvoton Technology Corporation America
June 2026

Joy of innovation
NUVOTON

| Agenda: Evaluation Pre-Requirement Steps

To start Nuvoton MCU Evaluation and Hands-on Training

You will need:

- Step1 Toolchain IDE – **Keil** (20min)
 - Step2 Nuvoton Tool USB Driver – **Nu-Link Driver** (5min)
 - Step3 Nuvoton BSP Package – **Sample Code Package** (5min)
 - Step4 Third-Party Terminal – **PuTTY** (5min)
 - Step5 Pre-requirement Verification (10min)
-
- Basic Need: Windows PC / NuMaker Board / USB cable
 - Additional Need: tool for dip switcher

Step 1: Toolchain IDE

- **Install** <KEIL IDE – MDK-ARM>

1

armKEIL

Home / Product Downloads

MDK-ARM
MDK-ARM Version 5.43a
Version 5.43a
MDK-ARM v5.xx is part of Keil MDK v6

Complete the following form to download the Keil software development tools.

Note
Please note that these installers do not include protection against tampering. To help ensure the integrity of your Arm software installer, we recommend that installers be executed from a trusted directory that contains only the installer.

Enter Your Contact Information Below

First Name: Pawan
Last Name: Ru
E-mail: pr@nuvoton.com
Company: Nuvoton Technology Corporation of America
Job Title: Principal Algorithm Developer
Country/Region: Select Your Country
State/Province: California
Phone: 14584741972

☐ Send me e-mail when there is a new update.
NOTICE: If you select this checkbox, you will receive an e-mail message from Nuvoton whenever a new update is available. If you don't wish to receive an e-mail notification, don't check this box.

Which device are you using?
(eg. STX552) **M0 / M23 / M4**

Arm will process your information in accordance with the Evaluation section of our [Privacy Policy](#).

☐ Please keep me updated on products, services and other relevant offerings from Arm. You can change your mind and unsubscribe any time.

Submit **Reset**

2
DONE

armKEIL

Home / Product Downloads

MDK-ARM
MDK-ARM Version 5.43a
Version 5.43a

- Review the hardware requirements before installing this software.
- Note the limitations of the evaluation tools.
- **MDK543A.EXE (872,548K)**
Thursday, August 21, 2025
- If you are evaluating the tools, be sure to request a quote for the full version of the tools.

- **Apply** <Keil MDK Nuvoton Edition – Full Cortex-M>

FAQ - Unable to successfully activate

This free service is available for use with Keil MDK Version 5.37 or later and only supports th

Submit

Nuvoton Technology Corp.

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Step 1: Toolchain IDE

- **Apply** <Keil MDK Nuvoton Edition – Full Cortex-M>

5

Keil MDK Nuvoton Edition – Full Cortex-M – Application Success

Thank you! Your application has been successfully submitted.

You will soon receive an automatic system email containing the Nuvoton License & Usage Statement for Keil MDK Nuvoton Edition–Full Cortex-M, and the activation guide.

If you have any questions, or if you did not receive the system-generated license email, please feel free to contact us at: SalesSupport@nuvoton.com.

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DONE

【Nuvoton License & Usage Statement】Keil MDK Nuvoton Edition – Full Cortex-M

Dear Mr./Ms. ,

Thank you for applying. Before using this service, please carefully read the following usage statement:

• If Nuvoton detects any suspicious behavior that violates Nuvoton [Terms of Use](#), [Privacy Policy](#), or related regulations on Nuvoton website (including other site regulations and relevant legal provisions), or service or the Keil MDK Nuvoton Edition – Full Cortex-M, or if other abnormal situations arise (such as unauthorized use of your account by a third party), Nuvoton Technology has the right to suspend or permanently terminate the related services and licenses.

• If you do not agree with the above statement, please stop using the services immediately and contact Nuvoton Technology to cancel your account.

• This license is valid from the time you receive this email until September 25, 2027.

You can now start legally using the Keil MDK Nuvoton Edition – Full Cortex-M service provided by Nuvoton Technology by following the activation steps below:

Activation Guide

1. Open Keil µVision as administrator.

☒

2. Go to the "File" menu, open "License Management" settings, switch to "User-Based License", and click on "Activate/Deactivate".

☒

3. Select the "License Server" option.

4. Enter the following URL: <https://nuvoton-free.licensing.keil.arm.com>

5. Press "Query" button and confirm if the "Keil MDK Nuvoton Edition – Full Cortex-M" appears.

6. Press "Activate" button.

☒

7. It will then display a successful activation message.

☒

If you are unable to open it properly, please refer to the [FAQs](#). If you have any questions or need assistance, please feel free to contact our customer support team at SalesSupport@nuvoton.com.

Thank you for using our service. We look forward to providing you with excellent service!

| Step 1: Toolchain IDE

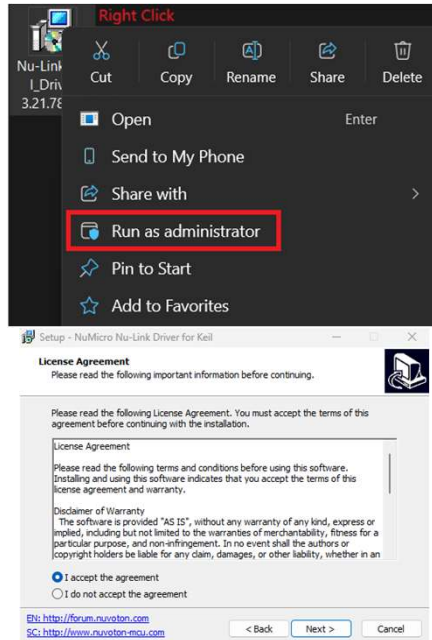
Other IDE supported by Nuvoton (Option)

- IAR EWARM
- NuEclipse (GCC) (Windows) - Free
- NuEclipse (GCC) (Linux) - Free

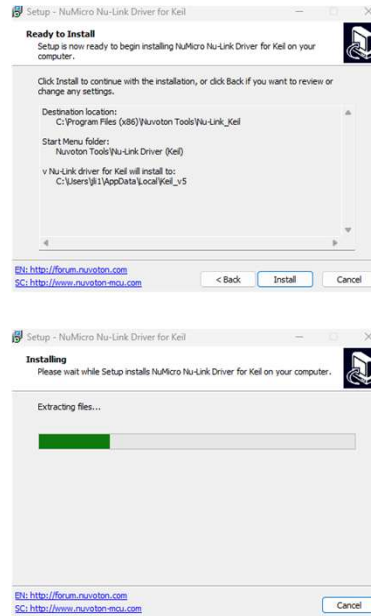
Step 2: Nuvoton Tool USB Driver

- **Install Nu-Link_Keil_Driver**

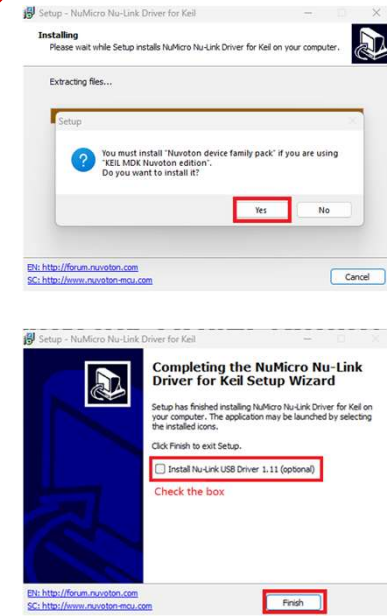
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| Step 2: Nuvoton Tool USB Driver

Other IDE Driver (Option)

- Download and install Nu-Link_IAR_Driver when using IAR EWARM.
- Skip this step when using NuEclipse.

Step 3: Nuvoton BSP Package

Method 1 – Nuvoton Website

- **Click** webpage link
- **Click 3. Board Support Package (BSP)**
- **Download** → unzip to your workspace (Keep in short directory path)
 - **M0: M031TC**
<https://www.nuvoton.com/board/numaker-m031tc/?index=2>
 - **M23: M2003FC**
<https://www.nuvoton.com/board/numaker-m2003fc/?index=2>

Method 2 – Github

- **Click** Github link
- **Download** → unzip to your workspace (Keep in short directory path)
 - **M0: M031TC**
<https://github.com/OpenNuvoton/M031BSP/archive/refs/heads/master.zip>
 - **M23: M2003FC**
<https://github.com/OpenNuvoton/M2003BSP/archive/refs/heads/master.zip>
 - **M4: M46x**
<https://github.com/OpenNuvoton/m460bsp/archive/refs/heads/master.zip>

Board Introduction

Board Features

Quick Start

1. Install the preferred toolchain. Please make sure at least one of the toolchains has been installed.

- [KEIL MDK Nuvoton edition M0 / M23](#)

- [IAR EWARM](#)

- [NuEclipse \(GCC\) \(Windows \)](#)

- [NuEclipse \(GCC\) \(Linux \)](#)

2. Download and install the latest Nuvoton Nu-Link Driver. Please install the Nu-Link USB Driver as well at the

- Download and install [Nu-Link Keil_Driver](#) when using Keil MDK.

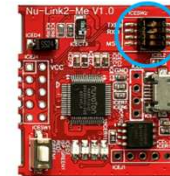
- Download and install [Nu-Link IAR_Driver](#) when using IAR EWARM.

- Skip this step when using NuEclipse.

3. Download and unzip the **Board Support Package (BSP).**

4. Hardware Setup

(1) Open the virtual COM (VCOM) function by changing Nu-Link2-Me VCOM Switch No. 1 and 2 to ON.



Step 4: Third-Party Terminal

- **Install PuTTY** <putty-64bit-0.83-installer.msi>
 - <https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html>

Download PuTTY: latest release (0.83)

[Home](#) | [FAQ](#) | [Feedback](#) | [Licence](#) | [Updates](#) | [Mirrors](#) | [Keys](#) | [Links](#) | [Team](#)
Download: [Stable](#) | [Snapshot](#) | [Docs](#) | [Privacy](#) | [Changes](#) | [Wishlist](#)

This page contains download links for the latest released version of PuTTY. Currently this is 0.83, released on 2025-02-08.

When new releases come out, this page will update to contain the latest, so this is a good page to bookmark or link to. Alternatively, here is a [permanent link to the 0.83 release](#).

Release versions of PuTTY are versions we think are reasonably likely to work well. However, they are often not the most up-to-date version of the code available. If you have a problem with this, already been fixed in those versions.

Package files

You probably want one of these. They include versions of all the PuTTY utilities (except the new and slightly experimental Windows pterm).

(Not sure whether you want the 32-bit or the 64-bit version? Read the [FAQ entry](#).)

We also publish the latest PuTTY installers for all Windows architectures as a free-of-charge download at the [Microsoft Store](#); they usually take a few days to appear there after we release them.

MSI ('Windows Installer')

64-bit x86:	putty-64bit-0.83-installer.msi	(signature)
64-bit Arm:	putty-arm64-0.83-installer.msi	(signature)
32-bit x86:	putty-0.83-installer.msi	(signature)

Unix source archive

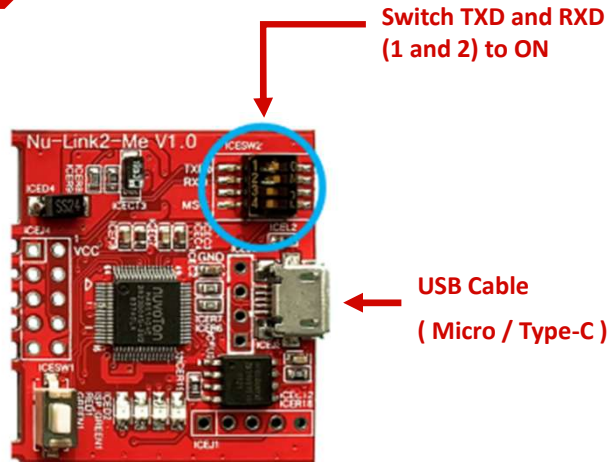
.tar.gz:	putty-0.83.tar.gz	(signature)
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- Note: If you already have preferred terminal, please skip this step.

Step 5: Verification – USB device COM Port

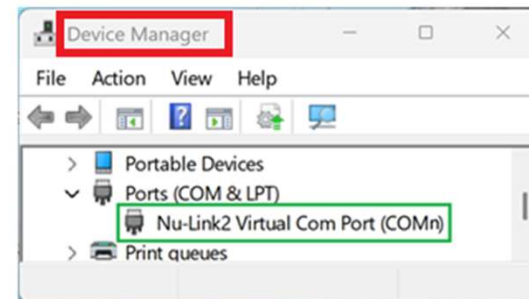
- Verify valid USB device in "Device Manager"

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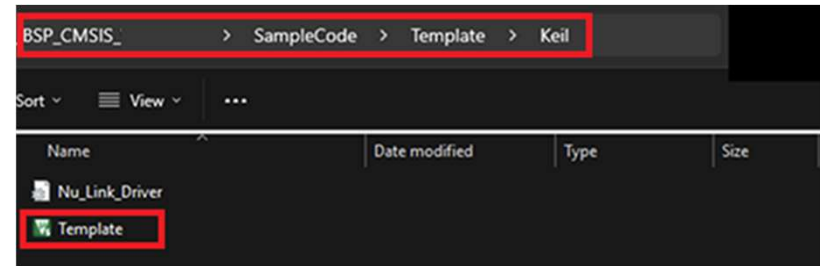
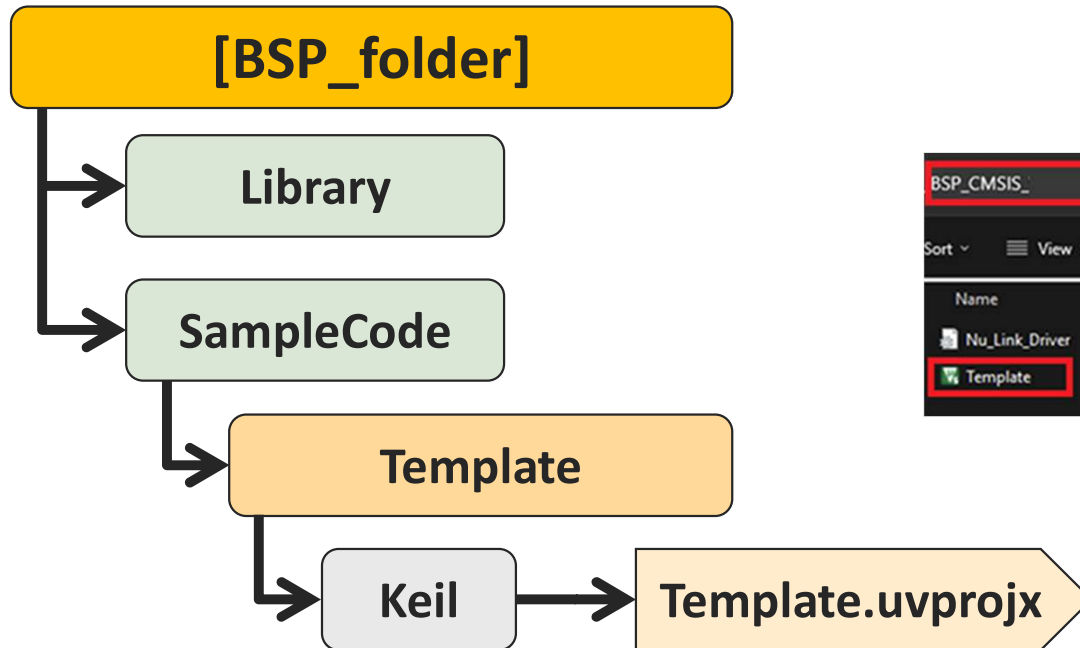
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DONE



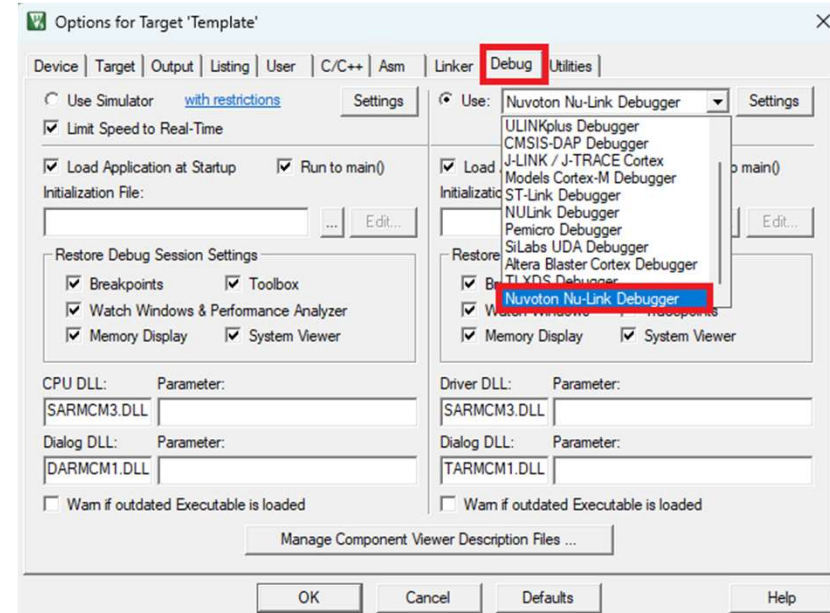
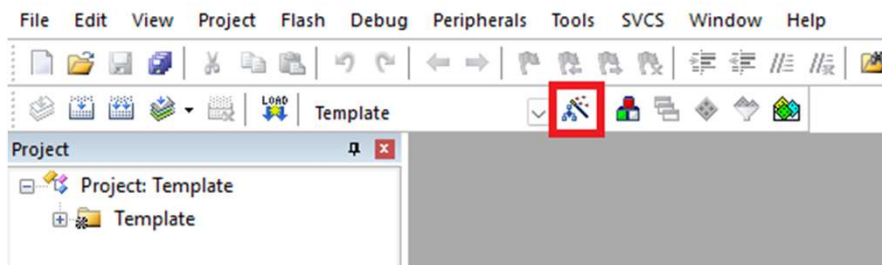
Step 5: Verification – Open Template

- Open "Template" project



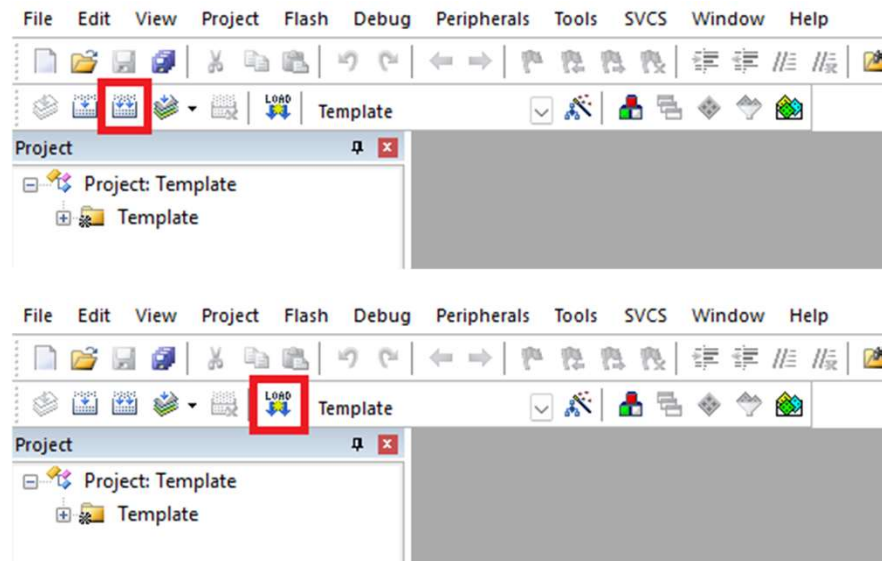
Step 5: Verification – Debugger Selection

- Select "Nuvoton Nu-Link Debugger" in "Options for Target"



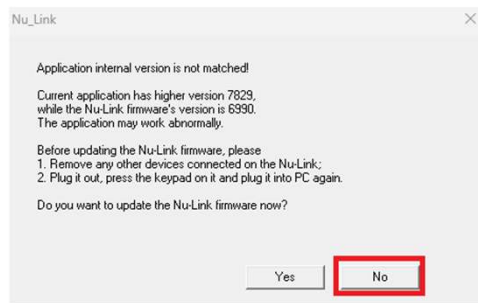
Step 5: Verification – Build & Download to EVB

- Click "Rebuild"
- Click "Download"

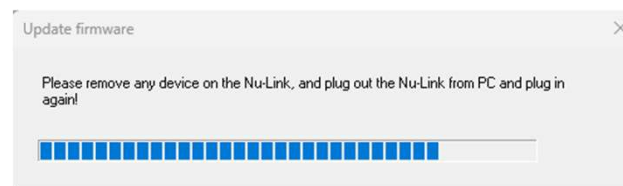
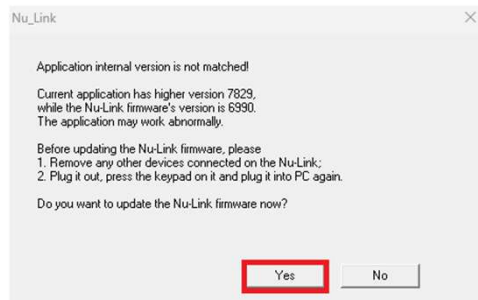


Step 5: Verification – Programmer Firmware

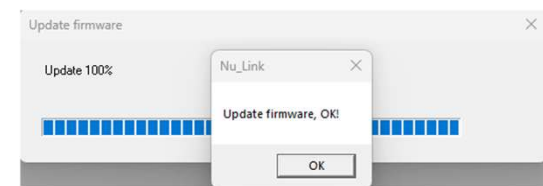
- Click "No" for quick skip (will pop up every time)



- Click "Yes" for programmer firmware update (no longer pop up)

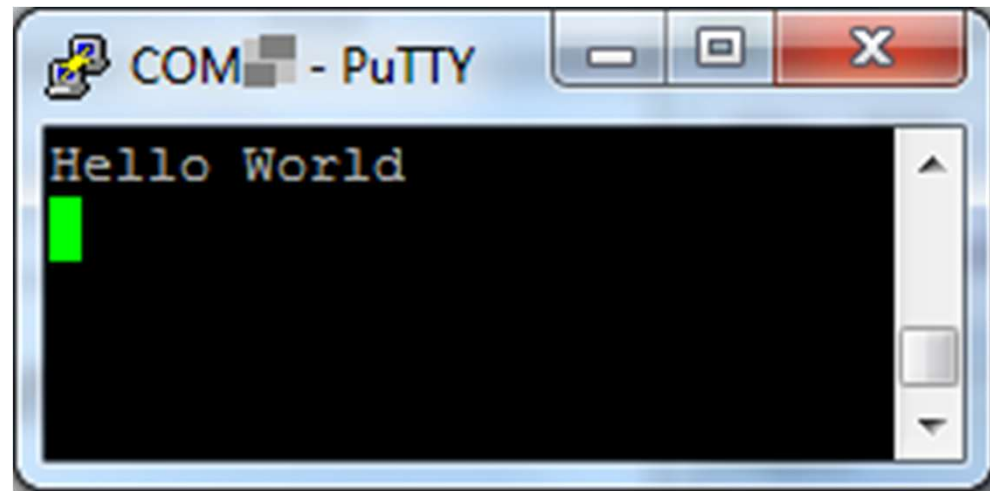
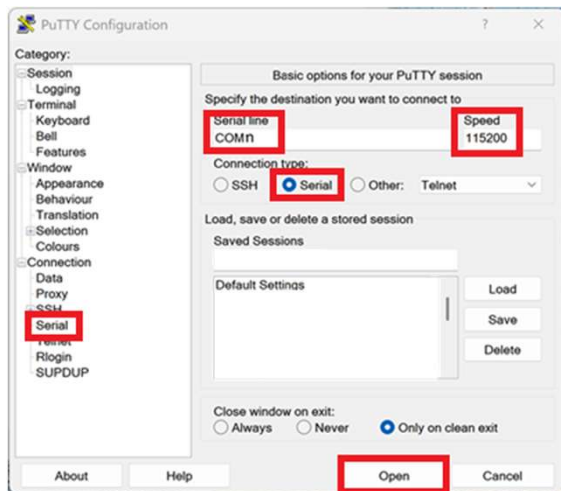


Unplug and re-plug in USB cable



Step 5: Verification – Hello World

- Open "PuTTY" or your preferred terminal
- Choose "Category – Connection – Serial "
- Key in "COMn" per Device Manager (Step 4)
- Speed - 115200 baud (8 bits - 1 stop bit - parity:none)
- Select "Serial" in Connection Type
- Click "Open"
- EVB - Press "nRESET" button
- Get "Hello World"



| Congratulations !

Thank you for completing Nuvoton evaluation pre-requirement steps.

Please contact Nuvoton MCU US tech support team in case you need further assistance.

MCU US Tech Support

MCU-US@nuvoton.com

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谢谢

謝謝

Děkuji

Bedankt

Thank you

Kiitos

Merci

Danke

Grazie

ありがとう

감사합니다

Dziękujemy

Obrigado

Спасибо

Gracias

Teşekkür ederim

Cảm ơn